Subject Description Form

Subject Code	ABCT1104
Subject Title	Biology in Daily Life
Credit Value	3
Level	1
Pre-requisite / Co-requisite/ Exclusion	Exclusion: ABCT1D04 or ABCT1101.
Objectives	This subject gives students an easy-to-understand introduction to how the human body works at the level of cells, tissues, organs and body systems. It also introduces common diseases, basic ideas of how drugs work, and a few modern biomedical technologies that affect health and society. The main aim of this subject is to arouse students' interest in biological developments so that they can appreciate the impact of biomedical science.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) have a basic understanding of biology princicples governing different life forms (b) have a basic understanding about human health and diseases (c) Appreciate recent developments in biotechnology and discuss their possible impact and ethical issues.
Subject Synopsis/ Indicative Syllabus	**1. Introduction to different life forms and Human Body** • What is life: from molecules to cells to a whole human body. • Overview of different groups of organisms relevant to human health: viruses, bacteria, protozoa, simple animals and humans. • Microbes in daily life. **2. From Cells to Body System** • Cells as the basic building blocks of life. • Introduction to various cell types such as bacteria, yeast and human cells • Simple structure and functions of major subcellular • Different major cell types in the human body • Cell division, growth and repair; abnormal growth and an introduction to cancer. • Organization of tissues, organs and systems in humans. • Homeostasis: how the body keeps internal balance **3. Genetics, heredity and human disease** • Genetic material: simple structure and role of DNA and RNA. • Genes and how they store information for making proteins. • From DNA to traits: basic idea of gene expression. • How genetic information is passed to offspring; examples of inherited traits.

- Introduction to genetic diseases and screening.

4. Infectious disease and immunity

- Types of pathogens: viruses, bacteria, protozoa and how they enter and spread in the body.
- Basic concepts of the immune response: innate and adaptive immunity, antibodies, vaccines.

5. Drugs and how they work (pharmacology basics)

- What is a drug; difference between drugs, biologics, medicines and poisons.
- How drugs act on the body explained in simple terms.
- Side effects, misuse of drugs, and basic concepts of safe medication use.

6. Modern biotechnology and human health

- Recent new drugs development and advanced biotechnology
- Impacts of biotechnology on everyday life, medicine and the environment.

Teaching/Learning Methodology

In the lectures, the basic concepts and knowledge will be delivered to the students. These knowledge and concepts will be further enhanced through tutorial exercises, discussions and debates during tutorials, and through assessments.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)		
		a	b	c
1. Classwork or Class participation	10%	✓	✓	✓
2. Mid Term Test 1	20%	✓	✓	
3. Mid Term Test 2	20%	✓	✓	
4. Final Exam	50%	✓	✓	✓
Total	100 %			

Classwork or class participation includes in-class tutorial questions which will be used to draw students' interest, improve their understanding and encourage participation in classroom discussion. Each student will also be required to read broadly and to complete a written assignment in which an understanding of some of the major concepts and knowledge has to be demonstrated. In this written assignment, a student will also need to express his/her critical evaluation of the impacts of a new development in biotechnology. This assignment will be in the form of a critical review essay.

A student will also need to take two tests (Mid term Test 1 & 2) which will gauge their learning outcomes at two separate stages of the subject. These assessments will also allow students to get feedbacks on their performance and how well they are achieving the learning outcomes.

There will also be an end of subject assessment which will assess all of the three learning outcomes. This will be in the form of an examination.

Student Study Effort Expected	Class contact:	
	▪ Lectures	34Hrs.
	▪ Tutorials	5Hrs.
	Other student study effort:	
	▪ Self Study	80Hrs.
	Total student study effort	119Hrs.
Reading List and References	**Eric J. Simon, Jean L. Dickey, Jane B. Reece** **Campbell Essential Biology with Physiology Fifth Edition** **Pearson 2014**	